

Ex. No: 5 Inter Process Communication (IPC)

PROGRAM 1: IPC USING PIPE

C PROGRAM

```
#include<stdio.h>

#include<stdlib.h>

#include<unistd.h>

#include<string.h>

#include<sys/wait.h>

int main()

{

int fd[2];

pid_t pid;

char message[] = "Hello from Child Process";

char buffer[100];

pipe(fd);

pid = fork();

if(pid == 0)

{

close(fd[0]);

write(fd[1], message, strlen(message)+1);

close(fd[1]);

exit(0);

}

else

{

wait(NULL);

close(fd[1]);

read(fd[0], buffer, sizeof(buffer));

printf("Message received from child: %s\n", buffer);

close(fd[0]);

}
```

```
return 0;
```

```
}
```

Output:

```
(kali㉿kali)-[~]  
$ gcc pipe.c -o pipe  
  
(kali㉿kali)-[~]  
$ ./pipe  
Message received from child: Hello from Child Process
```

SHELL SCRIPT:

```
#!/bin/bash
```

```
echo "Hello from Child Process" | cat
```

Alternative Shell Script

```
#!/bin/bash
```

```
(
```

```
echo "Message from Child Process"
```

```
) | while read msg
```

```
do
```

```
echo "Parent Received: $msg"
```

```
done
```

Output:

```
(kali㉿kali)-[~]  
$ chmod +x pipe.sh  
  
(kali㉿kali)-[~]  
$ ./pipe.sh  
Parent Received: Message from Child Process
```